

Department of Computer Science and Engineering

Assignment Questions

Course Code & Name : CSA15 - CLOUD COMPUTING AND BIG DATA ANALYTICS

Assignment Title: Design and Critical Analysis of a Cloud-Based Service Architecture for a Scalable Online Education

Organization

The screenshot displays a web application titled "Cloud-Based Online Education Service" running on localhost:8501. The application features a navigation sidebar on the left with the following options: Home (selected), Cloud Architecture, Virtualization Demo, Web Service Demo, IaaS / PaaS / SaaS / XaaS, Student Service, and Results. The main content area includes a title "Cloud-Based Online Education Service" with a subtitle "Simple implementation for CSA15 - Cloud Computing and Big Data Analytics". Below the title is a section titled "Online Education Cloud" with the text "This demo shows a simple cloud architecture for an online education organization." A table displays the following metrics:

Students	Courses	Virtual Machines	Availability Target
10,000	250	4	99.9%

Below the table is a section titled "Main Services" with the following list:

- **IaaS:** Virtual machines, storage and networking
- **PaaS:** Application development and deployment platform
- **SaaS:** Learning management, video meetings and collaboration
- **XaaS:** Combination of cloud services

A blue box at the bottom of the main content area contains the text: "The project is an educational simulation. It does not create real cloud resources."

Navigation

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Cloud Architecture

Virtualization Demo

Web Service Demo

IaaS / PaaS / SaaS / XaaS

Student Service

Results

Cloud-Based Online Education Service

Simple implementation for CSA15 - Cloud Computing and Big Data Analytics

Proposed Cloud Architecture

Students / Teachers

Internet / HTTPS

Web Application

PaaS / IaaS / SaaS / XaaS

Navigation

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Cloud Architecture

Virtualization Demo

Web Service Demo

IaaS / PaaS / SaaS / XaaS

Student Service

Results

Database

SaaS Services

Cloud Storage

Virtualization Layer

Hypervisor

Physical Cloud Server

VM 1: LMS | VM 2: Exams | VM 3: API | VM 4: DB

Why this architecture?

Virtual machines share physical hardware, while PaaS simplifies application deployment and SaaS provides ready-to-use educational tools. Cloud storage handles course files and videos.

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Cloud-Based Online Education Service

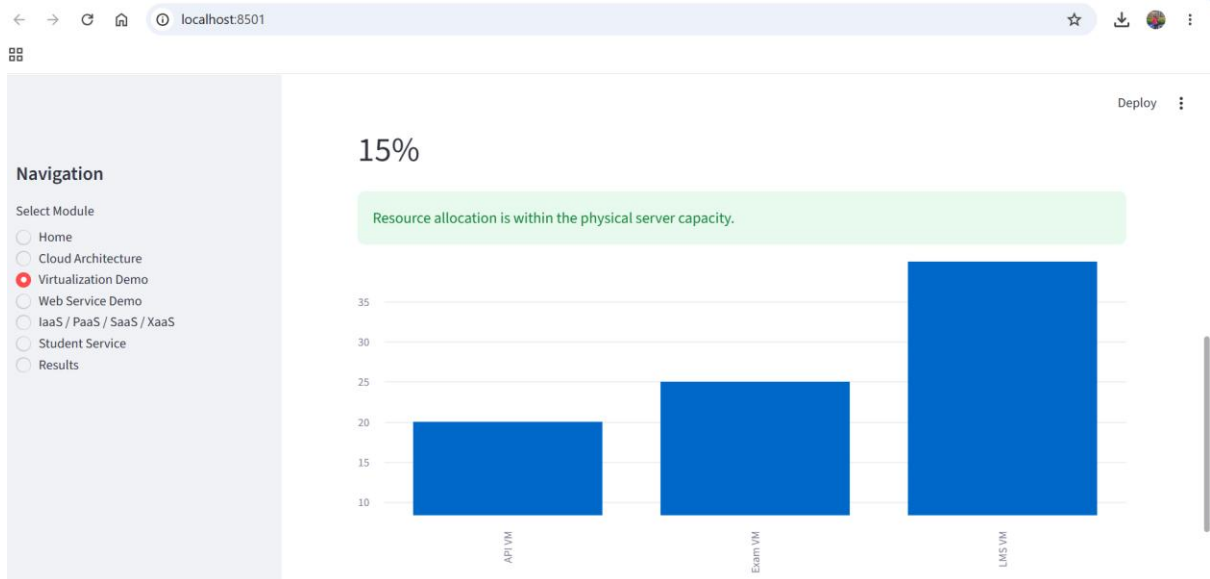
Simple implementation for CSA15 - Cloud Computing and Big Data Analytics

Server Virtualization Demonstration

Move the sliders to simulate resource allocation among virtual machines.

LMS VM CPU (%)

Exam VM CPU (%)



← → ↺ 🏠 ⓘ localhost:8501 ☆ ⬇️ 🌐 ⋮

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Deploy ⋮

Web Service / REST API Demonstration

A simple simulated API shows how applications exchange data over a network.

Student Name

Gayathri

Course

Cloud Computing

Send API Request

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Service

Simple implementation for CSA15 – Cloud Computing and Big Data Analytics

Cloud Service Model Comparison

	Model	Control	Management	Scalability	Cost	Example
0	IaaS	High	User manages OS/apps	High	Pay for resources	VM, storage, network
1	PaaS	Medium	Provider manages platform	High	Pay for platform/use	Develop and deploy apps
2	SaaS	Low	Provider manages almost everything	High	Subscription/use	Ready-to-use software
3	XaaS	Varies	Shared responsibility	Very High	Usage based	Combination of services

Recommended combination

Use IaaS for flexible infrastructure, PaaS for developing the education portal, SaaS for collaboration/communication, and XaaS to combine services.

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IaaS / PaaS / SaaS / XaaS

Student Service

Results

Simple Online Education Service

Student Name

Gayathri B.R

Email

Gaya@gmail.com

Enroll in

Cloud Computing

Enroll

Gayathri B.R enrolled successfully in Cloud Computing.

Navigation

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Cloud Architecture

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IaaS / PaaS / SaaS / XaaS

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Implementation Results

```
{  "Cloud architecture": "Implemented as a visual simulation"  "Virtualization": "VM resource allocation demonstrated"  "Web services": "REST-style JSON exchange demonstrated"  "IaaS/PaaS/SaaS/XaaS": "Compared and recommended"  "Education service": "Student enrollment demo implemented"  "Deployment": "Ready to run locally with Streamlit"}
```

Implementation completed successfully.